

1. **Consumption with Optimal Portfolio Choice.** **CRRA-RateRisk** shows that for a Merton (1969)-Samuelson (1969) consumer facing return $\log \mathfrak{R}_{t+1} \sim \mathcal{N}(\mathfrak{r} - \sigma_{\mathfrak{r}}^2/2, \sigma_{\mathfrak{r}}^2)$ on the only financial asset available, the optimal marginal propensity to consume is approximately

$$\kappa \approx \mathfrak{r} - \rho^{-1}(\mathfrak{r} - \vartheta) - (\rho - 1)(\sigma_{\mathfrak{r}}^2/2) \quad (1)$$

- a) Use this equation to discuss the parametric restriction(s) under which an increase in unavoidable financial risk $\sigma_{\mathfrak{r}}^2/2$ will cause a (precautionary) decline in consumption. Explain whether, under such parameter values, the income effect of an increase in $\mathfrak{r}$ outweighs the substitution effect, *and explain in words the intuition for both the income effect and the substitution effect.*

Answer:

The restriction is $\rho > 1$. Under this restriction, the income effect outweighs the substitution effect. For a discussion of the two effects, see **PerfForesightCRRA**.

But when the consumer can choose how much to invest in the risky asset, financial risk is not unavoidable. **Portfolio-CRRA** derives the portfolio share ς that an optimizing consumer will invest in a risky asset earning return $\log \mathbf{R}_{t+1} \sim \mathcal{N}(\mathbf{r} - \sigma_{\mathbf{r}}^2/2, \sigma_{\mathbf{r}}^2)$ so that $\log \mathbf{R} = \mathbf{r}$ (where the subscriptless version of a return factor denotes its expectation: $\mathbf{R} \equiv \mathbb{E}_t[\mathbf{R}_{t+1}]$ (see **LogMeanMPS**)).¹ The remaining proportion $(1 - \varsigma)$ of the portfolio earns a riskless return $\mathbf{r} = \log \mathbf{R}$,² and we write the expected return premium factor as $\Phi \equiv \mathbf{R}/\mathbf{R}$ with expected return premium $\phi \equiv \log \Phi = \mathbf{r} - \mathbf{r} = \log \mathbf{R}/\mathbf{R}$; optimal choice of ς yields a portfolio whose realized return factor is written $\mathfrak{R}$ (using the Gothic font)

$$\log \mathfrak{R} \equiv \log \mathbb{E}_t[\mathfrak{R}_{t+1}] = \varsigma \phi + \mathbf{r}, \quad (2)$$

and **Portfolio-CRRA** shows that under these circumstances the optimal risky portfolio share is well approximated by

$$\varsigma \approx \left(\frac{\phi}{\rho \sigma_{\mathbf{r}}^2} \right). \quad (3)$$

and that the variance of the return on the optimally-chosen risky portfolio is approximately

$$\sigma_{\mathfrak{r}}^2 = \varsigma^2 \sigma_{\mathbf{r}}^2. \quad (4)$$

- b) Show that the precautionary effect of rate-of-return risk on the precautionary contribution to the MPC, after taking account of optimal portfolio adjust-

¹Note the subtlety that the subscriptless version of a logarithmic return, e.g. $\mathbf{r}$, is *not* equal to the mean: $\mathbb{E}_t[\mathbf{r}_{t+1}] = \mathbf{r} - \sigma_{\mathbf{r}}^2/2$.

²The bold font is used for the risky return and the narrow font for the safe return.

ment, is

$$-(\rho - 1) \left(\frac{(\phi/\rho)^2}{2\sigma_r^2} \right) \quad (5)$$

and explain the intuition for the result that the size of the precautionary effect *shrinks* as the risk grows larger.

Answer:

Substituting from (3), the precautionary effect after taking account of optimal portfolio adjustment is

$$-(\rho - 1) (\sigma_r^2/2) = -(\rho - 1) \left(\frac{\overbrace{\varsigma^2 \sigma_r^2}^{\text{from (4)}}}{2} \right) \quad (6)$$

$$= -(\rho - 1) \left(\frac{(\phi/\rho)^2}{2\sigma_r^2} \right) \quad (7)$$

The way to understand this is to break down the response to the increase in riskiness into two components. The first is the direct precautionary effect examined in **CRRA-RateRisk**, which works in the way intuition suggests (more risk implies lower consumption); the second effect is the ‘portfolio rebalancing’ effect, examined in **Portfolio-CRRA**: An increase in riskiness makes the consumer choose to invest less in the risky asset. What (5) tells us is that the consumer’s ‘flight from risk’ is so effective in reducing riskiness (because the portfolio share ς enters as a *squared* term in (6)) that the riskiness of the *portfolio* actually *declines* as the riskiness of the risky asset increases. (This result was also derived in **Portfolio-CRRA**). Now the overall reduction in consumption coming through the precautionary term makes sense: Precautionary saving is less than before *because the consumer optimally chooses less exposure to risk* than before.

- c) Now use the foregoing results to show that the MPC κ can be rewritten in terms of exogenous parameters (including the riskfree interest factor r) as

$$\kappa \approx r - \rho^{-1}(r - \vartheta) + (\rho - 1) \left(\frac{(\phi/\rho)^2}{2\sigma_r^2} \right) \quad (8)$$

and use this equation to discuss whether the total effect of an increase in risk on consumption is positive or negative *and explain why this result comes about*.

Answer:

See **C-With-Optimal-Portfolio** for the result and the logic.

d) Now show that the effect of an increase in risk on saving is given by

$$\mathbf{r} - \kappa = \rho^{-1}(\mathbf{r} - \vartheta) + \left(\frac{\rho + 1}{2}\right) \left(\frac{\phi^2}{\sigma_{\mathbf{r}}^2 \rho^2}\right) \quad (9)$$

and *explain how this result relates to the results retained earlier*. Comment, in particular, on the role that our assumption that the consumer has no labor income plays in obtaining the conclusion (hint: What further effect of interest rates is omitted when the consumer has no labor income?)

Answer:

See the discussion in **C-With-Optimal-Portfolio**.

References

- MERTON, ROBERT C. (1969): “Lifetime Portfolio Selection under Uncertainty: The Continuous Time Case,” *Review of Economics and Statistics*, 50, 247–257.
- SAMUELSON, PAUL A. (1969): “Lifetime Portfolio Selection by Dynamic Stochastic Programming,” *Review of Economics and Statistics*, 51, 239–46.